Group 1

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## Question 3

### 1. Add Client

Sequence of Steps

1. The system receives the new client's name and address.
2. StoreSystem generates a unique client ID.
3. StoreSystem creates a new Client object using the client ID, name, and address.
4. The new Client is added to the system's collection of clients.
5. The system returns the result of the operation.

Design Choices

The StoreSystem coordinates the operation because it maintains the collection of clients. The client ID is generated by the system so that every client has a unique identifier. A new client begins with no WishlistItem objects because the client has not added any products to the wishlist yet.

Responsibilities

StoreSystem: Generates the client ID, creates the Client, and adds it to the client collection.

Client: Stores the client's ID, name, address, and associated wishlist items.

UI: Supplies the client information and displays whether the operation was successful.

### 2. Add Products

Sequence of Steps

1. The system receives the product name, quantity in stock, and sale price.
2. StoreSystem generates a unique product ID.
3. StoreSystem creates a new Product object.
4. The new Product is added to the system's collection of products.
5. The system returns the result of the operation.

6. Steps 1 to 5 are repeated until there are no more products to add.

Design Choices

Each product is created separately so that every product receives its own unique product ID. Repetition is used because the requirement allows multiple products to be added during the operation. The StoreSystem maintains the product collection, while each Product stores its own product information.

Responsibilities

StoreSystem: Generates product IDs, creates Product objects, and adds them to the product collection.

Product: Stores the product ID, name, quantity in stock, and sale price.

UI: Accepts information for each product, allows additional products to be entered, and displays the result.

**3. Add Products to a Client Wishlist**

Sequence of Steps

1. The system receives a clientID, productID, and desired quantity.
2. StoreSystem searches for the Client identified by the clientID.
3. If the client does not exist, the operation returns an error and no changes are made.
4. StoreSystem searches for the Product identified by the productID.
5. If the product does not exist, the operation returns an error and no changes are made.
6. The system checks the client's existing WishlistItem objects for the specified Product.
7. If the Product is not already represented in the wishlist, a new WishlistItem is created containing the Product and desired quantity.
8. If the Product already exists in the wishlist, the existing WishlistItem quantity is changed to the new quantity.
9. The system returns the result of the operation.
10. These steps are repeated for each Product being added to the client's wishlist.

Design Choices

StoreSystem first verifies that both the Client and Product exist so that an invalid ID cannot create an incorrect wishlist entry. A WishlistItem represents the association between a Client and a Product and stores the desired quantity. If the Product is already present, the existing quantity is replaced rather than creating a duplicate entry. This directly follows the requirement that an existing wishlist quantity be changed to the new quantity.

Responsibilities

StoreSystem: Finds the Client and Product, validates the IDs, and coordinates the wishlist operation.

Client: Maintains the collection of WishlistItem objects belonging to that client.

WishlistItem: Associates a Product with the quantity desired by the Client and updates that quantity when necessary.

Product: Stores the information describing the Product.

UI: Accepts the IDs and quantities and displays the result of the operation.

**4. Display All Clients With Details**

Sequence of Steps

1. The UI requests all clients from StoreSystem.
2. StoreSystem retrieves the collection of Client objects.
3. Each Client provides its client details.
4. The client information is returned to the UI.
5. The UI displays each Client's details on a separate line.

Design Choices

StoreSystem is responsible for maintaining and retrieving the collection of clients. Each Client is responsible for its own stored information. Display formatting remains the responsibility of the UI instead of being placed inside the business classes.

Responsibilities

StoreSystem: Retrieves the collection of Clients.

Client: Provides its stored client details.

UI: Displays each Client on a separate line.

**5. Display All Products With Details**

Sequence of Steps

1. The UI requests all products from StoreSystem.
2. StoreSystem retrieves the collection of Product objects.
3. Each Product provides its product details.
4. The product information is returned to the UI.
5. The UI displays each Product's details on a separate line.

Design Choices

StoreSystem manages the collection of Products, so it is responsible for retrieving them. Each Product maintains its own product information. The UI is responsible only for presenting the information to the user.

Responsibilities

StoreSystem: Retrieves the collection of Products.

Product: Provides its product ID, name, quantity in stock, and sale price.

UI: Displays each Product on a separate line.

**6. Given ClientID, Display All Products in Wishlist With Quantities**

Sequence of Steps

1. The system receives a clientID.
2. StoreSystem searches for the Client with that ID.
3. If the Client does not exist, the system returns an error.
4. If the Client exists, StoreSystem obtains the Client's WishlistItem collection.
5. For each WishlistItem, the system obtains the associated Product information and desired quantity.
6. The product and quantity information are returned to the UI.
7. The UI displays each Product and its desired quantity on a separate line.

Design Choices

The Client must first be verified because wishlist information belongs to a specific Client. WishlistItem is used because it connects a Product to the quantity requested by that Client. Product information remains inside Product rather than being duplicated inside the wishlist entry.

Responsibilities

StoreSystem: Finds the Client and coordinates retrieval of the wishlist information.

Client: Provides access to its collection of WishlistItem objects.

WishlistItem: Provides the Product reference and desired quantity.

Product: Provides the Product details.

UI: Displays each wishlist Product and its quantity.